XIAMEN UNIVERSITY MALAYSIA
厦門大學馬來西亞分校

EEE312 Week 10
Lab Practice

Implement the algorithm for the secant method to find the root of the following nonlinear equation:

$$f(x) = 3 \sin x - 5 \cos x, x \in [-1, 2]$$

Show the results at every iteration of the algorithm until the solution converges when $| x_{i+2} - x_{i+1} | < \varepsilon$, where ε =0.000001.

// Beginning of the C Source code

```
1   // secant.c
2   #include <stdio.h>
3   #include <math.h>
4
5   float nonlinearFunction(float x)
6   {
7       float y;
8       y = 3.0*sin(x) - 5.0*cos(x);
9       return y;
10  }
```

Line 5-9: Define the function for nonlinear function f(x):

$$f(x) = 3 \sin(x) - 5 \cos(x)$$

Function main()

```
12  int main()
13  {
14      float x0 = 2.0; // Initialize x0 for the first iteration i = 0
15      float x1 = -1.0; // initialize x1 for the first iteration i = 0
16      float x2 = 0.0; // store the next iteration value x2
17      float f_x0 = 0.0; // store the function f(x) evaluated at x0
18      float f_x1 = 0.0; // store the function f(x) evaluated at x1
19      float error = 0.0; // store the error=fabs(x2-x1)
20      float epsilon = 0.000001; // stop the iteration if
21                                // error < epsilon
```

Line 14: Declare and initialize x0= 2.0.

Line 15: Declare and initialize x1= -1.0.

Line 16: Declare and initialize x2= 0.0.

Line 17: Declare a variable f_x0 to store the value of the function f(x0).

Line 18: Declare a variable f_x1 to store the value of the function f(x1).

Line 19: Declare a variable to store the error= | x_{i+2} - x_{i+1} |

Line 20: Declare a variable to store ε = 0.000001.

```
23      // secant method starts here ...
24      do {
25          f_x0 = nonlinearFunction(x0); // calculate f(x0)
26          f_x1 = nonlinearFunction(x1); // calculate f(x1)
27
28          x2 = (x0*f_x1 - x1*f_x0)/(f_x1-f_x0); // update x2
29
30          printf("\nx0=%f, x1=%f, x2=%f,", x0, x1, x2);
```

Line 25: Evaluate f(x)at x0value by calling nonlinearFunction().

Line 26: Evaluate f(x)at x1value by calling nonlinearFunction().

Line 28: Update the new value of x2using the secant method equation

$$x_2 = \frac{x_0 f(x_1) - x_1 f(x_0)}{f(x_1) - f(x_0)}$$

```
32          error = fabs(x2-x1); // calculate the difference between x2 and x1
33          printf("\nError=%f", error);
34
35          x0 = x1; // store the current x1 as the next x0
36          x1 = x2; // store the current x2 as the next x1
37      } while(error>epsilon);
38      printf("\nThe root of the nonlinear equation is x = %f.\n", x2);
39
40      return 0;
41  }
```

Line 32: Calculate the error using the formula error = | x_{i+2} - x_{i+1} |

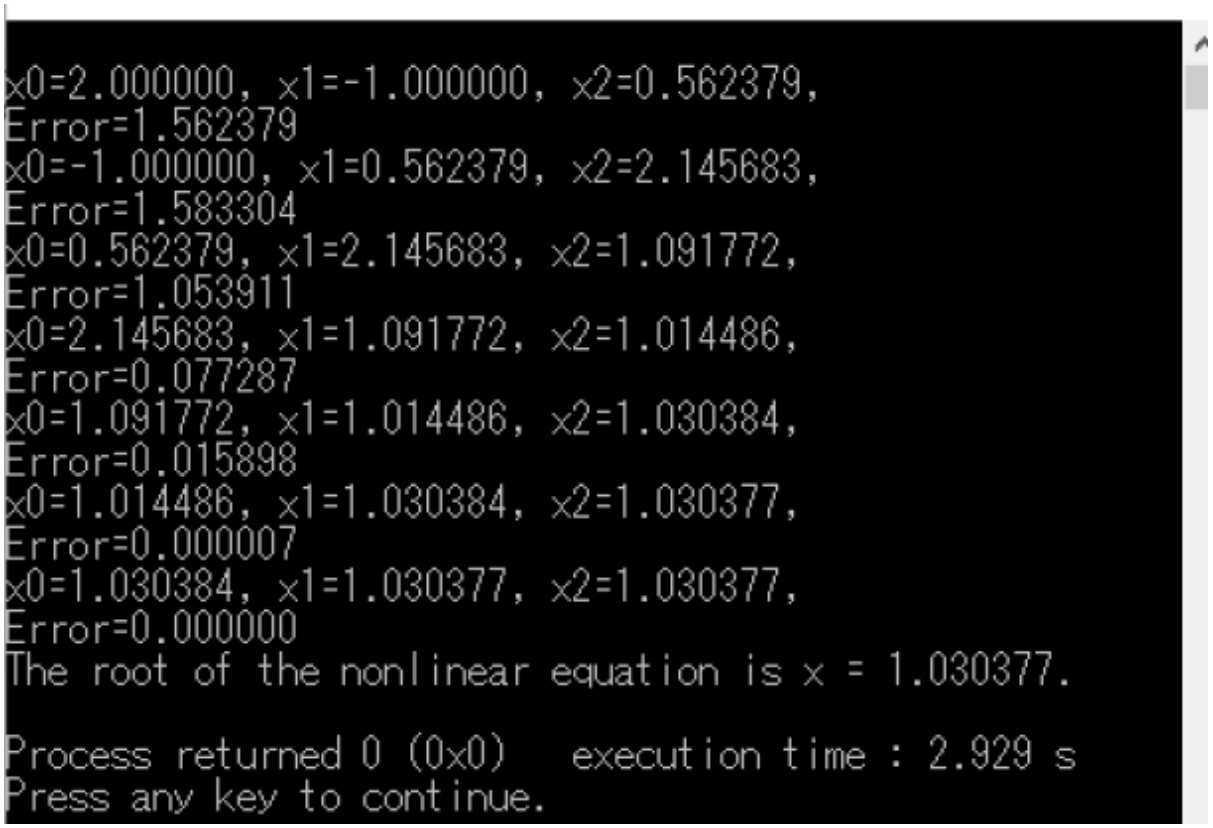
Line 33: Print out the error value.

Line 35: Store the updated x_1 in the current x_0 for the next iteration.

Line 36: Store the updated x_2 in the current x_1 for the next iteration.

Line 37: Check if error > epsilon. If yes, return to the beginning of the do-while loop and repeat the algorithm again. Otherwise, escape from the loop and output the final solution in Line 38.

Output


x0=2.000000, x1=-1.000000, x2=0.562379,
Error=1.562379
x0=-1.000000, x1=0.562379, x2=2.145683,
Error=1.583304
x0=0.562379, x1=2.145683, x2=1.091772,
Error=1.053911
x0=2.145683, x1=1.091772, x2=1.014486,
Error=0.077287
x0=1.091772, x1=1.014486, x2=1.030384,
Error=0.015898
x0=1.014486, x1=1.030384, x2=1.030377,
Error=0.000007
x0=1.030384, x1=1.030377, x2=1.030377,
Error=0.000000
The root of the nonlinear equation is x = 1.030377.
Process returned 0 (0x0) execution time : 2.929 s
Press any key to continue.